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Breaking the Nyquist Limit in Seismic Wavefield Modeling Using Implicit DRPSM Scheme

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Abstract

Accurate and efficient seismic wavefield modeling near or beyond the Nyquist limit remains a critical challenge in computational geophysics. In our previous work, we introduced an Implicit Finite Difference (IFD) method for solving the acoustic wave equation, which achieves high numerical accuracy with compact stencils. In this study, we further extend the method by integrating ideas from the Dispersion-Relation Preserving Stereo-Modeling (DRPSM) scheme proposed by Jing et al. (2017), which demonstrated that forward modeling is feasible using fewer than two grid points per wavelength. We propose a new hybrid approach, the Implicit DRPSM (IDRPSM) scheme, which combines the compactness and stability of IFD with the dispersion control of DRPSM to enable sub-Nyquist modeling. Through rigorous dispersion-relation analysis, we demonstrate that the IDRPSM method can significantly relax spatial sampling requirements while maintaining wavefield fidelity. Although numerical simulations in complex models such as BP 2D and homogeneous domains are ongoing, the theoretical results provide strong evidence for the feasibility of high-accuracy modeling with reduced grid density. The proposed IDRPSM scheme opens a promising pathway for efficient large-scale seismic modeling and imaging in high-frequency regimes.

Introduction

The continued growth of computational power has catalyzed a shift in seismic imaging from ray-based to wave-equation-based techniques. In methods such as reverse time migration (RTM) and full waveform inversion (FWI), the efficiency and accuracy of wave equation solvers are critical, as they constitute the computational backbone of these techniques. Despite advances in hardware, many state-of-the-art imaging and inversion workflows still rely on traditional numerical methods to solve the two-way wave equation in heterogeneous media on high-performance computing (HPC) systems.

The cost of solving the two-way wave equation increases significantly with higher frequencies and longer offsets, which are increasingly used to probe deeper and more complex subsurface targets. Thus, improving computational efficiency remains a major focus in both academia and industry (Liu, 2020b; Miao and Zhang, 2020; Suha et al., 2020). Two general strategies are commonly pursued: implementing established algorithms on modern parallel hardware such as GPUs (Liu et al., 2010, 2012; Suha et al., 2020), and designing new algorithms that improve both numerical efficiency and physical accuracy—the latter being the focus of our research.

Among existing schemes, the explicit finite-difference time-domain (FDTD) method is widely used due to its simplicity and locality. However, the method's accuracy and stability are tightly coupled to both temporal and spatial discretization. Time step size is constrained by the CFL condition and dispersion errors, while spatial grid spacing is even more critical, as the total computational cost scales roughly with the inverse cube of grid spacing in 3D problems. Although pseudo-spectral methods can achieve high accuracy with coarser grids (up to half of the minimum wavelength), the 3D Fast Fourier Transform (FFT) remains computationally expensive for large -scale simulations. As a result, high-order explicit finite-difference (EFD) schemes (Dablain, 1986; Fornberg, 1988) continue to dominate production codes, especially when implemented with cache optimization techniques for CPU or shared memory on GPU platforms (Liu et al., 2010, 2012; Suha et al., 2020). However, EFD schemes suffer from the "saturation effect" (Kosloff et al., 2010): increasing the order of the operator yields diminishing returns in accuracy due to fundamental limitations of the FD formulation. Although coefficient optimization (He et al., 2019; Miao and Zhang, 2020) can alleviate this to some extent, it cannot eliminate the effect entirely.

Implicit finite-difference (IFD) methods offer an alternative, achieving comparable or better accuracy with much smaller spatial stencils (Zhou and Zhang, 2011). These schemes have long been employed in one-way wave equation migrations using various techniques, such as standard finite differences (Ma, 1983; Claerbout, 1985; Li, 1991) and Fourier finite-difference (FFD) methods (Ristow and Rühl, 1994). Their application to two-way wave equations has been explored by multiple authors (Lele, 1992; Ashcroft and Zhang, 2003; Zhou et al., 2008; Kosloff et al., 2010; Zhou and Zhang, 2011; Liu, 2013; Liu, 2020b). A key challenge in applying IFD schemes is the need to solve multi-diagonal (banded) systems of equations, typically via LU decomposition, which can be computationally demanding. To address this, Zhou and Zhang (2011) proposed pre-factored compact schemes using L+U decomposition, avoiding full matrix inversion.

In our previous work, we introduced an explicit formulation of IFD schemes by leveraging multi-time causal and anti-causal integrations inspired by anti-alias filtering techniques used in Kirchhoff migration (Lumley et al., 1994). With proper coefficient optimization, our method achieved spectral-like resolution while remaining fully explicit and memory-efficient. The computational overhead was limited to approximately 50% more than a standard 8th-order EFD method, but without requiring matrix inversion or additional memory. This made the method suitable for modern parallel architectures while preserving the accuracy advantages of implicit schemes.

Building upon that foundation, this paper explores the integration of dispersion-relation preserving stereo-modeling (DRPSM) techniques originally proposed by Jing et al. (2017), which demonstrated that forward modeling can be performed using fewer than two points per wavelength—breaking the traditional Nyquist constraint. Inspired by this concept, we propose a hybrid approach, termed the **Implicit DRPSM (IDRPSM)** method, which combines the compactness and stability of IFD with the dispersion-preserving properties of DRPSM.

In this paper, we present the theoretical formulation and dispersion analysis of the IDRPSM method. Our results demonstrate that the proposed scheme can achieve accurate wavefield simulation below the Nyquist limit. While synthetic tests in homogeneous and complex models are planned as future work, the dispersion analysis confirms that IDRPSM is a promising candidate for efficient, high-resolution seismic modeling in bandwidth-limited or resource-constrained applications.

Theory and Method

In exploration geophysics, seismic wave propagation is governed by the acoustic wave equation:

$$\frac{\partial^2 p}{\partial t^2} = v^2\left(\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 p}{\partial z^2}\right) + s(x, y, z, t), \tag{1}$$

where $p(x, y, z, t)$ is the pressure wavefield, $v(x, y, z)$is the propagation velocity, and $s(x, y, z, t)$ is the source term. Among numerical methods used to solve Equation (1), the explicit finite-difference time-domain (FDTD) scheme is one of the most widely adopted due to its simplicity and locality. To reduce dispersion errors from time discretization, one may use either smaller time steps or higher-order schemes (Zhou and Zhang, 2011). Alternatively, the errors introduced by temporal and spatial discretization can be jointly minimized through optimization, which aims to reduce the mismatch between numerical and analytical phase velocities (Etgen, 2007; Etgen and Brandsberg-Dahl, 2009).

In this paper, we adopt the Lax-Wendroff fourth-order time discretization (Liu and Luo, 2019; Liu, 2020a) across all schemes. Our primary focus is on optimizing spatial discretization. For clarity, we present derivations and dispersion analysis using the second derivative in the x-direction, with other directions handled analogously.

To analyze and compare the numerical performance of different wavefield modeling schemes, we examine the dispersion relations of four representative methods: EFD, IFD, the dispersion-relation preserving stereo-modeling (DRPSM) method (Jing et al., 2017), and the newly proposed hybrid scheme, IDRPSM. The DRPSM approach showed that forward modeling is possible beyond the classical Nyquist limit, using fewer than two grid points per wavelength. Building on this idea, our proposed IDRPSM method combines th e compactness and stability of IFD with the sub-Nyquist capability of DRPSM. For each scheme, we derive the corresponding numerical dispersion relation and apply coefficient optimization to minimize phase velocity errors over a target frequency band. This unified analysis provides a clear assessment of each method's ability to preserve wave propagation fidelity, particularly under coarse-grid (sub- Nyquist) conditions.

Explicit Finite Difference (EFD)

The explicit finite-difference (EFD) scheme approximates spatial derivatives using a symmetric stencil composed of weighted differences of neighboring grid points. For the second spatial derivative in the xxx-direction, the general 2N-point EFD formulation can be written as:

$$D_i^2 = \sum_{n=1}^{N} a_n \frac{(u_{i+n} - 2u_i + U_{i-n})}{(nh)^2} \tag{2}$$

where u_i represents the wavefield value at grid point i, h is the spatial grid spacing, and a_n are the finite-difference coefficients associated with each pair of symmetric stencil points located n grid points away from the center. This form is often referred to as the generalized high-order central difference approximation. When N=1, Equation (2) reduces to the classic second-order central difference scheme. For higher accuracy, especially in the simulation of high-frequency wavefields, larger stencils (larger NNN) with optimized coefficients a_n are used.

$$\theta'(\theta) = \sum_{n=1}^{N} a_n (1 - cos(n\theta)). \tag{3}$$

We can measure the accuracy of the compact method in equation 2 from a Fourier analysis of the spatial derivative. For this purpose, we set $\theta = kh$ being the modified wavenumber, and θ^2 is the true wavenumber for the second-order derivatives. The numerical approximation $\theta'(\theta)$ in equation 3 of second-order derivative can be obtained by applying Fourier analysis on equation 2.

To improve the accuracy of the EFD scheme, the finite-difference coefficients a_n in Equation (2) are optimized to minimize numerical dispersion. Rather than relying solely on Taylor series expansion, we adopt a dispersion-relation-based optimization strategy. The goal is to reduce the phase velocity mismatch between the numerical and exact solutions across a specified range of normalized wavenumbers. This is achieved by minimizing the following objective function:

$$E = \int_{\theta_{min}}^{\theta_{max}} \left(\theta'(\theta) - \theta^2\right)^2 \xi(\theta) d\theta, \tag{4}$$

where $\xi(\theta)$ is the weighting function, The value of the modified wavenumber is limited in the range of

$$0 \le \theta_{min} \le \theta \le \theta_{max} \le \pi. \tag{5}$$

The weighting function $\xi(\theta)$) allows for frequency-dependent emphasis during optimization, and the integration bounds θ_{min} and θ_{max} define the target wavenumber band of interest, typically within

By minimizing the objective function EEE, we obtain a set of coefficients a_n that yield minimal dispersion error over the desired spectral range. This optimization framework enhances the phase accuracy of the EFD method, particularly in cases where coarse spatial sampling is used near or below the Nyquist limit. The accuracy of the scheme can be quantitatively evaluated by comparing the numerical and exact dispersion relations. Two common metrics are used: the absolute dispersion error and the relative dispersion error, defined respectively as:

$$E_{abs}(\theta) = \theta'(\theta) - \theta^2, \tag{6.1}$$

$$E_{rel}(\theta) = \frac{\theta'(\theta) - \theta^2}{\theta^2} \times 100\%. \tag{6.2}$$

Implicit Finite Difference (IFD)

The general form of the IFD stencil used in this study is given by:

$$D_i^2 + \sum_{m=1}^{M} b_m (D_{i+m}^2 + D_{i-m}^2) = \sum_{n=1}^{N} a_n \frac{(u_{i+n} - 2u_i + u_{i-n})}{(nh)^2} \tag{7}$$

where D_i^2 represents the second spatial derivative at grid point i, h is the grid spacing, and a_n, b_m are the finite-difference coefficients for the explicit and implicit parts of the stencil, respectively. The left-hand side introduces coupling between neighboring second derivatives, forming a compact banded system that characterizes the implicit nature of the method. When all $b_m = 0$, Equation (7) reduces to the standard explicit finite-difference formulation.

To assess and optimize the accuracy of the IFD scheme, we derive the numerical dispersion relation corresponding to Equation 7 as:

$$\theta'(\theta) = \frac{2\left(\sum_{n=1}^{N} a_n \left(1 - cos(n\theta)\right)\right)}{1 + 2\sum_{m=1}^{M} b_m \, cos(m\theta)}. \tag{8}$$

As with the EFD method, we determine the optimal IFD coefficients $\{a_m, a_m\}$ by minimizing the phase error over a target wavenumber range using the dispersion error functional defined in Equation (4). This process ensures that the numerical phase velocity remains as close as possible to the true velocity, even for coarse spatial sampling. The same absolute and relative error metrics introduced earlier (Equation 6) are used to evaluate the numerical accuracy of the optimized IFD scheme. The primary advantage of the IFD approach is its ability to achieve spectral-like accuracy using minimal stencils, making it highly suitable for large-scale simulations where both memory efficiency and numerical precision are essential.

Figure 1 compares the numerical dispersion curves of various finite-difference schemes with the analytical (theoretical) solution. The solid blue curve represents the theoretical dispersion relation, while the dashed curves correspond to standard explicit finite-difference (EFD) schemes of increasing order (8th to 64th), and the solid curves represent different implicit finite-difference (IFD) stencils, including 3–2, 4–1, 4–2, and 5–1 formulations. Panel (a) presents the full-range comparison from low to high normalized phase values, while panel (b) provides a zoomed-in view of the high-frequency region, where dispersion errors become more significant.

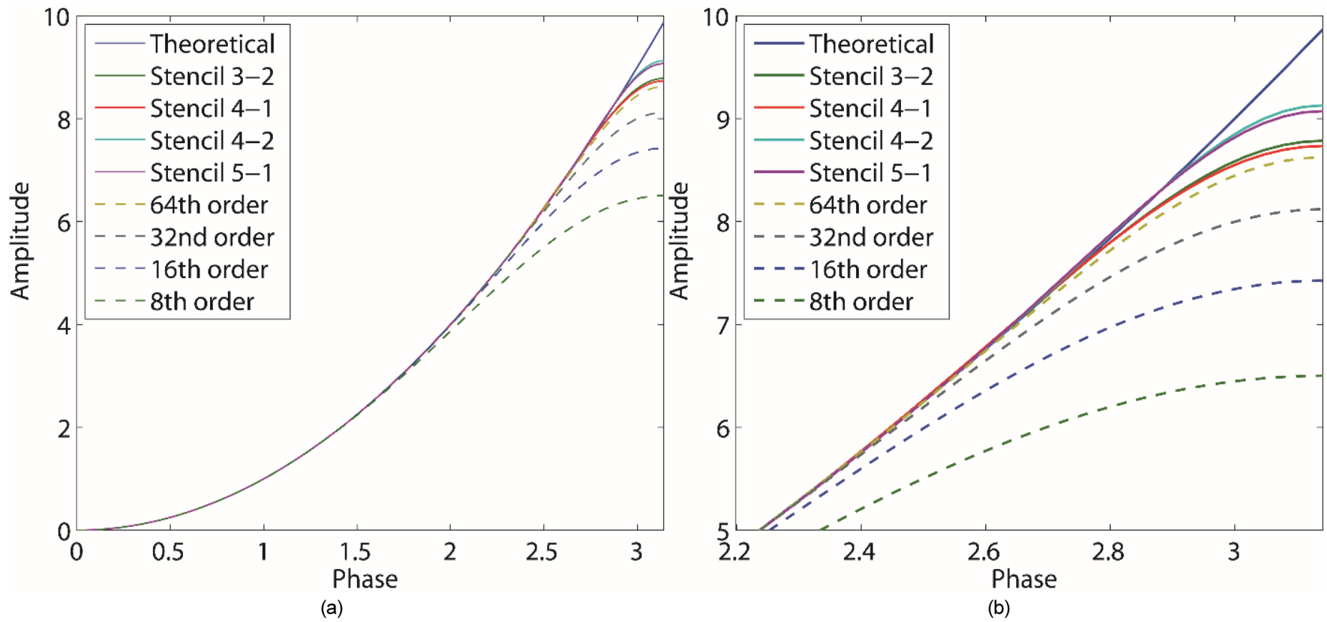


Figure 1—Dispersion curves for theoretical solution (solid blue curve), implicit (solid curves), and standard EFD methods (dashed curves). Figure (a) is the comparison from low to high frequencies, Figure (b) is the zoom-in of (a) focused on the high-frequency zone.

The results clearly demonstrate that IFD schemes provide significantly better phase accuracy than EFD counterparts, especially in the high-wavenumber regime. Even compact IFD stencils (e.g., 4–2 or 5–1) outperform much higher-order EFD methods (e.g., 32nd and 64th order), closely following the theoretical dispersion curve. This confirms that IFD methods are more effective in preserving wavefield fidelity, particularly for simulations targeting high-frequency content or operating under coarse spatial discretization near the Nyquist limit.

Dispersion-relation preserving stereo-modeling (DRPSM)

The Dispersion-Relation Preserving Stereo-Modeling (DRPSM) method, introduced by Jing et al. (2017), aims to accurately replicate wave dispersion behavior using fewer than two grid points per wavelength —effectively breaking the classical Nyquist limit. This is achieved by carefully constructing numerical approximations that match the exact dispersion relation over a wide frequency band, allowing for stable and accurate modeling even under coarse spatial sampling.

Unlike traditional second-order or higher-order schemes that only involve second derivatives, the DRPSM method introduces both second- and first-order spatial derivatives into the approximation of the Laplacian operator. Its general form is given by:

$$D_i^2 + \sum_{m=1}^{M_1} a_{m_1}\frac{u_{i+m_1} - 2u_i + u_{i-m_1}}{h^2} + \sum_{m_2=1}^{M_2} b_{m_2}\frac{u'_{i+m_2} - u'_{i-m_2}}{h} \qquad (9)$$

where u_i is the wavefield at grid point i, h is the spatial grid spacing, and u'_i represents the first-order spatial derivative of the wavefield at point i. The coefficients a_{m_1} and b_{m_2} are determined through optimization to minimize the dispersion error in equation 11 across a target range of normalized wavenumbers.

$$\theta'(\theta) = 2\sum_{m_1=1}^{M_1} a_{m_1}\left(1-\cos\left(m_1\theta\right)\right) + 2\theta\sum_{m_2=1}^{M_2} b_{m_2}\sin\left(m_2\theta\right). \tag{10}$$

$$J = \int_0^{\theta_{max}} \left(\theta^2 - \theta'\left(\theta\right)\right)^2 d\theta. \tag{11}$$

$$\frac{1}{2}\frac{\partial J}{\partial a_{m_1}} = \int_0^{\theta_{max}} F\frac{\partial F}{\partial a_{m_1}} d\theta = 2\int_0^{\theta_{max}} F(1-\cos(m_1\theta))d\theta = 0, \tag{12.1}$$

$$\frac{1}{2}\frac{\partial J}{\partial b_{m_2}} = \int_0^{\theta_{max}} F\frac{\partial F}{\partial b_{m_2}} d\theta = -2\int_0^{\theta_{max}} F\theta\sin(m_2\theta)\, d\theta = 0. \tag{12.2}$$

Following the original implementation in Jing et al. (2017), the DRPSM scheme is constructed with matching stencil widths for the second- and first-order derivative terms, i.e., $M_1 = M_2$. The optimization of the dispersion relation is carried out over extended wavenumber ranges, progressively increasing θ_{max} as the stencil length grows. Specifically, the authors selected $\theta_{max} = 1.0\pi$, 1.1π, 1.2π, 1.34π, 1.36π corresponding to $M_1 = M_2 = 2,3,4,5,6$, respectively. This approach enables the DRPSM scheme to preserve accurate dispersion behavior well beyond the Nyquist frequency, particularly when longer stencils are used.

Figure 2 compares the dispersion characteristics of various numerical schemes for approximating the second spatial derivative. Panel (a) shows the numerical dispersion curves plotted against the inverse of the grid sampling ratio 1/G1/G1/G, where GGG denotes the number of grid points per wavelength. The theoretical solution is shown in blue, while numerical curves are provided for the IFD 4–2 scheme, 64th-order EFD, and DRPSM schemes with stencil lengths M=2M = 2M=2 to M=6M = 6M=6. All DRPSM coefficients are adopted from Jing et al. (2017). Panel (b) presents the absolute dispersion error relative to the theoretical solution. It is evident that IFD 4–2 achieves superior accuracy in the mid-frequency range, while DRPSM schemes provide increasingly improved accuracy at high wavenumbers as MMM increases. The 64th-order EFD method, despite its wide stencil, exhibits larger errors due to saturation effects. Panel (c) shows the relative error percentage, which highlights a known limitation of DRPSM schemes: large relative errors in the low- frequency band. This is due to the small denominator θ2\theta^2θ2 used in the relative error definition, which amplifies even small absolute errors at long wavelengths. Nonetheless, all DRPSM schemes maintain sub-percent-level errors across a wide frequency band, indicating excellent high-frequency performance. Overall, this comparison demonstrates the strengths and trade-offs among different methods: the IFD scheme excels in balanced accuracy and compactness, while DRPSM achieves sub-Nyquist dispersion control at the cost of low-frequency precision in relative terms.

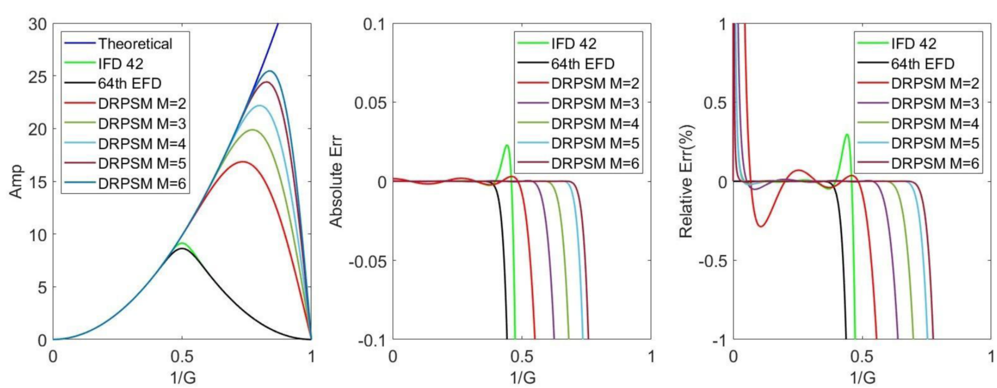


Figure 2—Comparison of dispersion relations of different approximation of second derivatives. (a) is the comparison between theoretical and numerical solutions; (b) is the absolute errors between the FD schemes and exact solution shown in (a); (c) is the relative errors between the FD schemes and exact solution shown in (a). Note the for the DRPSM schemes, the parameters are from Jing's paper (2017).

In summary, the DRPSM scheme enhances numerical accuracy by combining traditional second-derivative terms with dispersion- correcting gradient terms, enabling effective wavefield simulation at reduced grid density. This theoretical foundation is instrumental in our development of the IDRPSM scheme, which integrates the DRPSM concept into a compact implicit framework.

Implicit Dispersion-relation preserving stereo-modeling (IDRPSM)

To combine the accuracy and compactness of implicit finite-difference (IFD) schemes with the sub-Nyquist dispersion control of the dispersion-relation preserving stereo-modeling (DRPSM) approach, we propose a hybrid method: Implicit DRPSM (IDRPSM). This method aims to achieve high fidelity in wavefield simulation while using fewer grid points per wavelength, thus reducing computational cost and memory usage.

The IDRPSM scheme incorporates both symmetric and antisymmetric derivative terms, similar to DRPSM, but introduces an implicit formulation through coupling of multiple neighboring second derivatives. The general form of the scheme is:

$$D_i^2 + \sum_{n=1}^{N} c_n \left(D_{i+n}^2 + D_{i-n}^2\right) = \sum_{m_1=1}^{M_1} a_{m_1} \frac{u_{i+m_1} - 2u_i + u_{i-m_1}}{h^2} + \sum_{m_2=1}^{M_2} b_{m_2} \frac{u'_{i+m_2} - u'_{i-m_2}}{h}. \tag{13}$$

To evaluate and optimize the accuracy of the IDRPSM scheme, we derive its numerical dispersion relation by assuming a harmonic wave solution of the form $u(x) = e^{i\theta x/h}$. Substituting this into Equation (13) yields the following closed-form expression for the numerical squared wavenumber:

$$\theta'(\theta) = \frac{2\left(\sum_{m_1=1}^{M1} a_{m1}\left(1-\cos(m_1\theta)\right) + \theta \sum_{m_2=1}^{M2} b_{m2}\sin(m_2\theta)\right)}{1+2\sum_{n=1}^{N} c_n\cos(n\theta)}. \tag{14}$$

where $\theta = kh$ is the normalized wavenumber. The numerator combines both symmetric and antisymmetric contributions from the spatial stencil, while the denominator captures the implicit coupling effect.

To obtain the optimal coefficients $\{a_{m_1}, a_{m_2}, c_n\}$, we minimize the squared phase velocity error across a chosen frequency band. The optimization objective function is defined as:

$$J(a, b, c, \theta_{max}) = \int_0^{\theta_{max}} \left(\theta^2 - \theta'(\theta)\right)^2 \xi(\theta)\, d\theta. \tag{15}$$

subject to the constraint:

$$0 \le \theta_{max} \le 2\pi. \tag{16}$$

where θ_{max} determines the upper limit of the optimization band. The weighting function $\xi(\theta)$ is chosen to normalize the impact of the denominator in Equation (14) and is given by:

$$\xi(\theta) = \left(1 + 2\sum_{n=1}^{N} c_n\cos(n\theta)\right)^2 \tag{17}$$

This weighting ensures that the contribution of each frequency component is balanced with respect to the implicit amplification introduced by the denominator. By minimizing the functional J, the IDRPSM scheme achieves optimal phase accuracy over the specified frequency range while retaining compactness and numerical stability. The inclusion of both symmetric and antisymmetric terms in a rational dispersion form allows IDRPSM to capture wave physics more accurately, especially when operating near or below the Nyquist limit.

To numerically solve the optimization problem defined in Equation (15), we discretize the integration interval [0, θ_{max}] using $N\theta$ equally spaced wavenumber samples θ_j, and reformulate the minimization as a linear least-squares problem. The optimal coefficient vector c is obtained by solving the normal equations:

$$c = \left(A^T A\right)^{-1} A^T b. \tag{18}$$

where $c \in \mathbb{R}^{m_1+M_2+N}$ contains all unknown coefficients:

$$c = \left(a_1, \dots a_{M_1}; b_1, \dots, b_{M_2}; c_1, \dots, c_N\right)^T \tag{19.1}$$

he matrix $A \in \mathbb{R}^{N\theta \times \left(M_1+M_2+N\right)}$ is formed by concatenating column vectors derived from the discrete dispersion relation (Equation 14). It consists of three blocks:

The first M_1 columns correspond to the symmetric second-derivative terms:

$$Am_1 = \left(2\left(1 - \cos\left(m_1\theta_j\right)\right)\right)_{N\theta\times 1}, \quad 1 \le i \le M_1. \tag{19.2}$$

The next M_2 columns correspond to the antisymmetric first-derivative terms:

$$Am_2 = 2\theta_j \sin\left(m_2\theta\right)_{N\theta\times 1}, \quad M_1 + 1 \le i \le M_1 + M_2. \tag{19.3}$$

The final N columns correspond to the implicit (denominator) cosine terms:

$$A_n = -2\theta_j^2 \cos\left(n\theta_j\right)_{N\theta\times 1}, \quad M_1 + M_2 + 1 \le i \le M_1 + M_2 + N. \tag{19.4}$$

The target vector b $\in \mathbb{R}^{N\theta}$ represents the exact dispersion values:

$$b = \left(\theta_j^2\right)_{N\theta\times 1}, \tag{19.5}$$

This least-squares formulation efficiently yields the optimal coefficient set that minimizes the dispersion error across the specified frequency band. Once solved, the coefficients can be used to construct the IDRPSM operator in Equation (10) for high-accuracy, sub- Nyquist modeling.

IDRPSM only one term in c_N, specially, G=1.47 points/ wavelength for M1=M2=2, N=1.

In our implementation of the IDRPSM scheme, we adopt a simplified formulation with a single implicit term, i.e., N=1, to balance efficiency and accuracy. As with the DRPSM scheme, we extend the optimization range θ_{max} progressively with increasing stencil size. Specifically, we select θ_{max} = 1.0π, 1.2π, 1.3π, 1.4π, 1.5π for corresponding configurations. In the case where M1=M2=2 and N=1N = 1, the optimized IDRPSM operator achieves accurate modeling with as few as 1.47 grid points per wavelength, demonstrating its ability to operate below the classical Nyquist threshold. This result confirms the potential of the IDRPSM method for high-resolution, memory-efficient seismic modeling in bandwidth-constrained scenarios.

Figure 3 compares the dispersion characteristics of various second-derivative approximation schemes when only one implicit term is used (N=1) in the IDRPSM formulation. Panel (a) shows the dispersion curves of the numerical solutions relative to the theoretical curve (blue), plotted against the inverse of the grid sampling ratio 1/G. The figure includes the IFD 4–2 scheme, the 64th-order EFD scheme, and several IDRPSM schemes corresponding to different stencil lengths M=2 to M=6, with M1=M2=M. All IDRPSM configurations utilize a single denominator term, maintaining the simplicity of the operator.

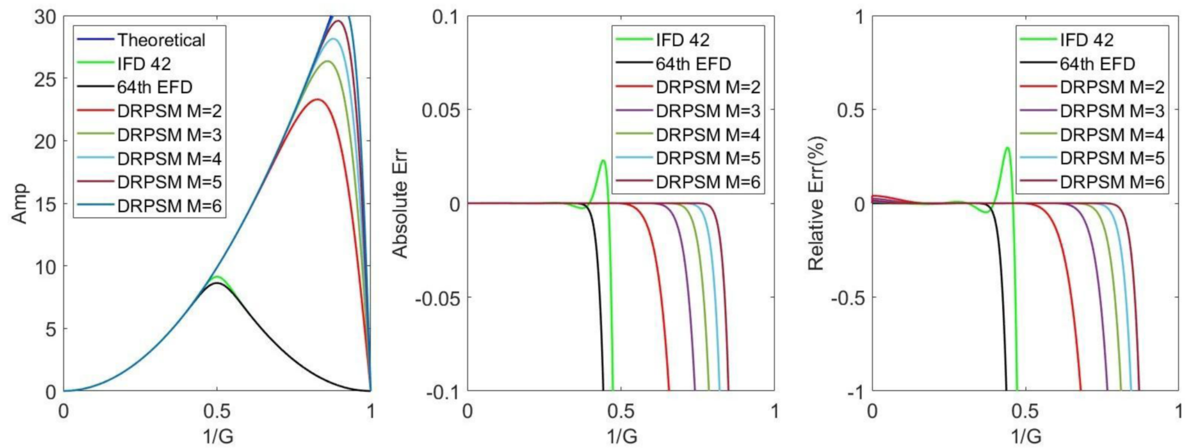


Figure 3—Comparison of dispersion relations of different approximation of second derivatives, only one term for IFD term (N=1). (a) is the comparison between theoretical and numerical solutions; (b) is the absolute errors between the FD schemes and exact solution shown in (a); (c) is the relative errors between the FD schemes and exact solution shown in (a).

Panel (b) displays the absolute error between the numerical and theoretical dispersion curves, while panel (c) shows the corresponding relative error in percentage. The results demonstrate that, even with only one implicit term, the IDRPSM schemes achieve excellent agreement with the theoretical dispersion relation up to high frequencies, significantly outperforming the 64th-order EFD method. Notably, IDRPSM with M=2 already enables accurate wavefield representation with as few as 1.47 grid points per wavelength, validating its sub-Nyquist modeling capability. These results confirm that the IDRPSM scheme can achieve high accuracy and spectral fidelity with a compact stencil and minimal implicit structure, making it well-suited for efficient seismic wavefield simulation under coarse spatial discretization.

Table 1 presents the optimized coefficients $\{a_m\}$, $\{b_m\}$, $\{c_n\}$ used in the IDRPSM schemes for stencil sizes M=2 through M=6, with a single implicit term (N=1). The values are obtained by minimizing the dispersion error functional defined in Equation (15), following the methodology outlined in this section. As the stencil length increases, more terms are activated and gradually tuned, allowing the dispersion relation to better match the theoretical curve over an extended frequency band.

Table 1—Optimized finite-difference coefficients for the IDRPSM scheme with increasing stencil length M=M1=M2=2 to 6 and one implicit term (N=1). Coefficients {a_m} correspond to symmetric second-derivative terms, {b_m} to antisymmetric first-derivative terms, and c_1 to the single implicit coupling term.

a_1	**3.8159111931**	**4.1996752623**	**4.4171152853**	**4.5744436617**	**4.6945067104**
a_2	-0.110905842	-0.254084212	-0.348902227	-0.418909045	-0.4716352491
a_3	0	-0.007471055	-0.020977472	-0.0355504411	-0.0489916350
a_4	0	0	-0.000947543	-0.0035720620	-0.0074058609
a_5	0	0	0	-0.0002086026	-0.0009911245
a_6	0	0	0	0	-7.642768e-05
b_1	-1.502684834	-1.558267840	-1.561937120	-1.5552662307	-1.5456529173
b_2	0.021679162	0.0728718475	0.1200782566	0.16273698968	0.20028048096
b_3	0	0.0016640413	0.0070962056	0.01543185107	0.02554385117
b_4	0	0	0.0002211107	0.00131825995	0.00366337016
b_5	0	0	0	5.1511591e-05	0.00040104421
b_6	0	0	0	0	2.0500862e-05
c_1	-0.2732725	-0.34951803	-0.39084390	-0.4197297084	-0.4411773558

Synthetic data tests

To evaluate the numerical accuracy and dispersion characteristics of the proposed schemes, we present two synthetic examples. The first test involves wave propagation in a homogeneous acoustic medium, allowing a clear comparison of waveform symmetry and numerical dispersion across different numerical schemes under controlled conditions. The second example uses the BP 2D velocity model, which features complex geological structures such as salt bodies and sharp velocity contrasts. This more realistic scenario demonstrates the performance of each scheme in capturing wavefront behavior and phase fidelity in challenging subsurface environments.

Figure 4 demonstrates the forward modeling results of a point source located at the center of a 4 km × 4 km acoustic domain using four numerical schemes. The subfigures (a–d), arranged in anti-clockwise order starting from the top-left, correspond to: (a) Implicit Finite Difference (IFD) 4-2 scheme, (b) 16th-order Explicit Finite Difference (EFD), (c) 32nd-order EFD, and (d) Pseudo-spectral (PS) scheme. All simulations are based on the acoustic wave equation with identical modeling parameters: peak frequency of 18 Hz, time step Δt=2 ms, total simulation time t=10 s, and spatial grid spacing h=20 m. Free-surface boundary conditions are applied on all sides to minimize artificial reflections during the long simulation duration.

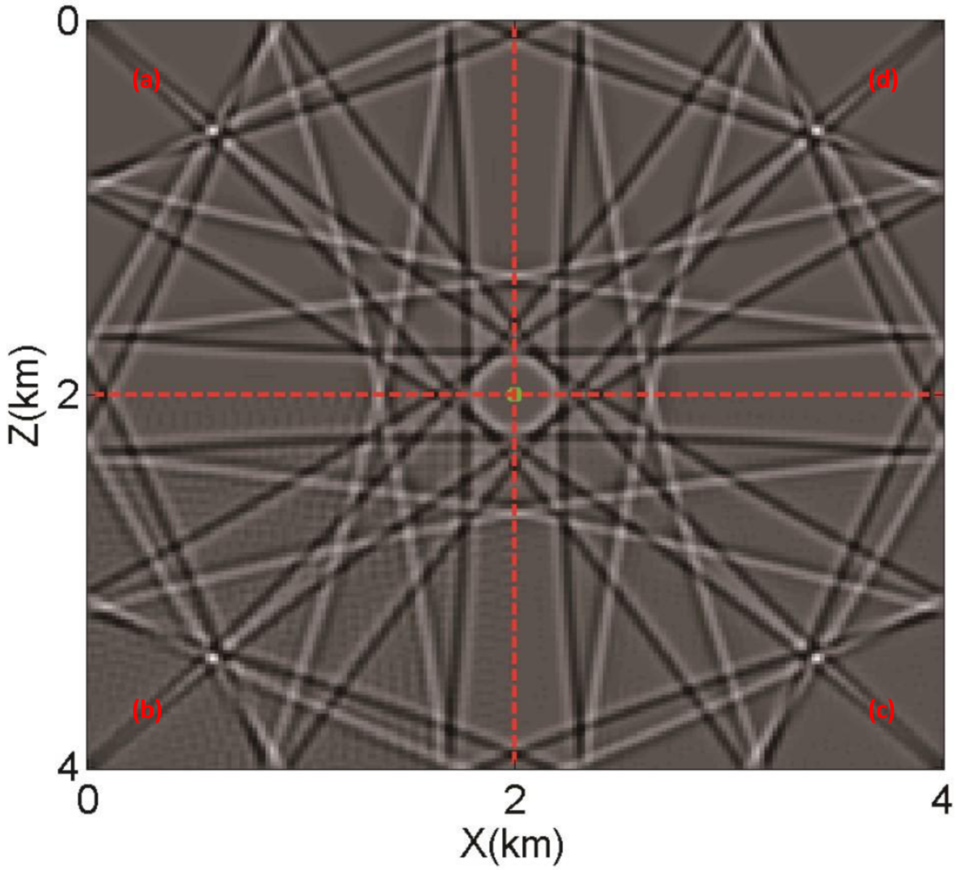
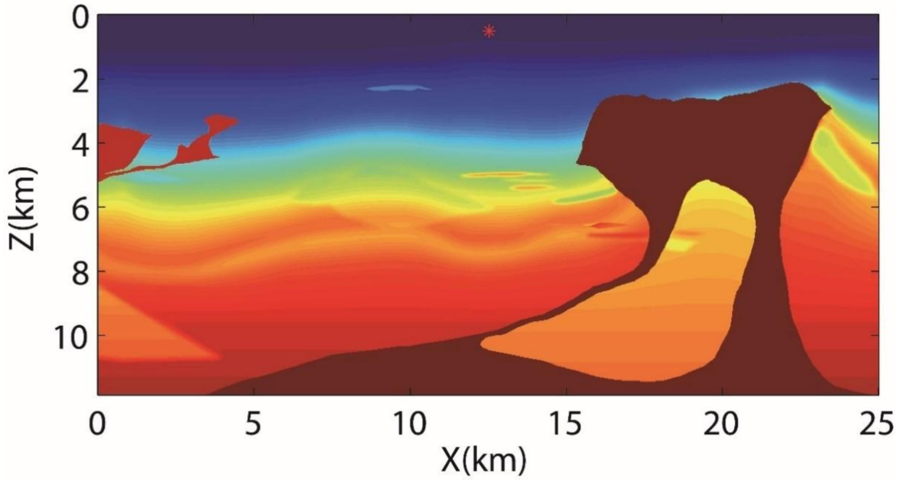


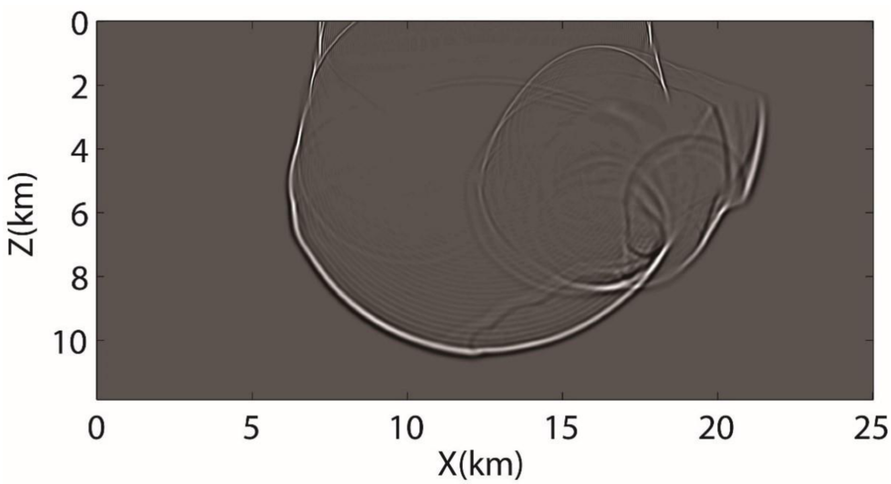
Figure 4—Forward modeling results of a point source located at the center of a 4 km × 4 km acoustic medium using four different numerical schemes. Subfigures (a) to (d), arranged in an anti-clockwise order starting from the top-left, correspond to: (a) Implicit 4-2 scheme, (b) Explicit Finite Difference (EFD) 16th-order scheme, (c) EFD 32nd-order scheme, and (d) Pseudo-spectral scheme.

The red dashed lines mark the source location at the model center. While all methods produce outward-propagating wavefields, the IFD 4-2 scheme demonstrates significantly improved numerical accuracy over both 16th- and 32nd-order EFD schemes, particularly in terms of reduced dispersion and cleaner wavefront symmetry. Remarkably, the IFD result closely matches the pseudo-spectral solution despite using a more compact stencil, highlighting its effectiveness for long-time wavefield modeling with high phase fidelity and computational efficiency.

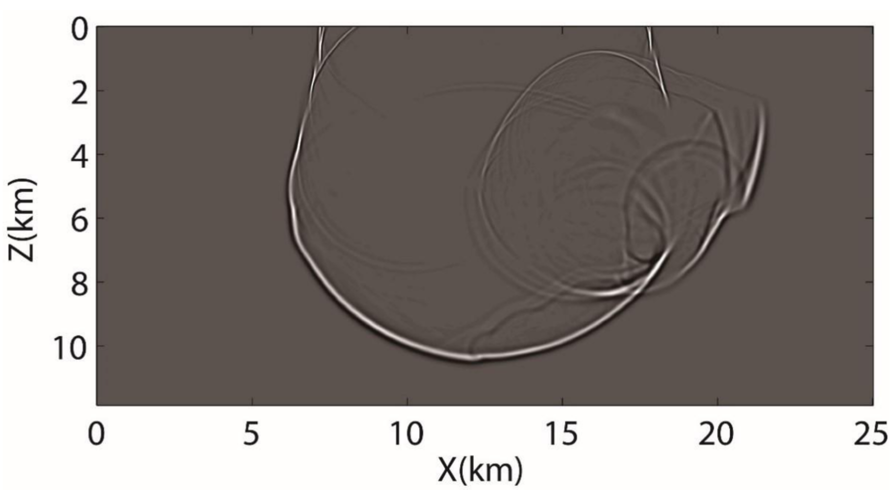
Figure 5 (a) is the BP 2D velocity model with complex geological structures, including high-contrast salt bodies and low-velocity zones. A point source (red asterisk) is located near the surface to simulate acoustic wave propagation. Panels (b) and (c) display wavefield snapshots at 3.6 s simulated using (b) the 32nd-order Explicit Finite Difference (EFD) scheme and (c) the Implicit Finite Difference (IFD) 4-2 scheme, respectively. Both simulations employ a peak frequency of 12 Hz, time step Δt=3.5 ms, and spatial grid spacing Δx=Δz=25 m.



(a)



(b)



(c)

Figure 5—Acoustic wavefield modeling in the complex BP 2D velocity model. (a) The BP 2D velocity model, featuring salt bodies, velocity contrasts, and structural complexity. A point source is placed near the surface (red asterisk). (b) Wavefield snapshot at 3.6 s using a 32nd-order Explicit Finite Difference (EFD) scheme; (c) same snapshot using the Implicit Finite Difference (IFD) 4-2 scheme.

While both methods capture the overall wave propagation pattern, the IFD result demonstrates notably reduced numerical dispersion compared to the EFD scheme, particularly in deeper regions and around complex interfaces. Despite its relatively compact stencil, the IFD scheme achieves accuracy comparable to that of the pseudo-spectral method (not shown), which is widely regarded as a gold standard for minimizing dispersion errors. This highlights the superior phase fidelity and modeling capability of the IFD approach, making it well-suited for high-fidelity simulations in complex geological settings.

Conclusions

In this study, we proposed a new numerical wavefield modeling method, the Implicit Dispersion-Relation Preserving Stereo-Modeling (IDRPSM) scheme, which integrates the compactness and numerical stability of implicit finite-difference (IFD) methods with the sub- Nyquist dispersion control of the DRPSM framework. This hybrid formulation is built on our previous work on explicit IFD schemes and is inspired by the DRPSM methodology introduced by Jing et al. (2017), which demonstrated the feasibility of accurate forward modeling below the Nyquist limit.

We derived the IDRPSM operator as a rational function of trigonometric terms and formulated a least-squares optimization framework to compute the coefficients by minimizing phase velocity errors over a specified frequency range. Dispersion analysis showed that the IDRPSM scheme outperforms high-order explicit finite-difference (EFD) and standard IFD methods, achieving excellent phase fidelity with significantly reduced stencil size. Notably, the configuration with M=2 and N=1 supports stable and accurate wavefield propagation using only 1.47 points per wavelength.

It is important to note that while the theoretical formulation and dispersion validation of the IDRPSM scheme are completed in this manuscript, the full synthetic tests and wavefield modeling results based on IDRPSM will be presented in future work. These tests will evaluate the method's performance in both homogeneous and complex geological models under realistic seismic acquisition setups.

The proposed IDRPSM scheme offers a promising pathway for high-resolution and memory-efficient modeling in large-scale seismic applications such as reverse time migration (RTM) and full waveform inversion (FWI), especially in bandwidth-limited or hardware- constrained environments.

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